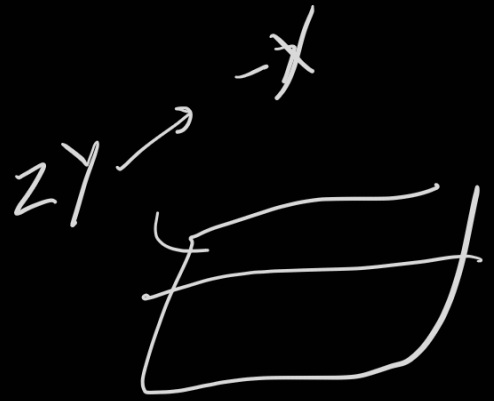


$$R_y = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

1) a)

$$R_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos & -\sin \\ 0 & \sin & \cos \end{bmatrix}$$



$$\text{Reflection} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow M = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & 0 & -\frac{\sqrt{3}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{3}}{2} & 0 & \frac{1}{2} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & 0 & -\frac{\sqrt{3}}{2} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} & 0 & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2\sqrt{2}} & -\frac{1}{\sqrt{2}} & \frac{1}{2\sqrt{2}} \\ \frac{\sqrt{3}}{2\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{2\sqrt{2}} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \frac{1}{2} & 0 & -\frac{\sqrt{3}}{2} & 2 \\ \frac{\sqrt{3}}{2\sqrt{2}} & -\frac{1}{\sqrt{2}} & \frac{1}{2\sqrt{2}} & 1 \\ \frac{\sqrt{3}}{2\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{2\sqrt{2}} & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix} = T$$

$2\frac{\sqrt{3}}{\sqrt{2}} \times 2$

1) (5)

$$P' = T P$$

$$T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 4 - \sqrt{3}/2 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 4 \\ -2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 + \frac{4\sqrt{3} + 5}{2\sqrt{2}} \\ -3 + \frac{4\sqrt{3} - 3}{2\sqrt{2}} \\ 1 \end{bmatrix}$$

$$O' = T \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -3 \\ 1 \end{bmatrix}$$

For points in the axis of Rotation we know

$$Rn = n$$

$$(R - I)n = 0$$

$$\begin{bmatrix} -\frac{1}{2} & 0 & -\frac{\sqrt{3}}{2} \\ \frac{-\sqrt{3}}{2\sqrt{2}} & \frac{1}{\sqrt{2}} - 1 & -\frac{1}{2\sqrt{2}} \\ \sqrt{3} & 1 & 1 - 1 \end{bmatrix} \begin{bmatrix} n \\ y \\ z \end{bmatrix} = 0$$

$$\begin{bmatrix} 2\sqrt{2} & \sqrt{2} & 2\sqrt{2} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\underline{x + \sqrt{3}z = 0} \quad (i)$$

$$\frac{z}{\sqrt{2}} + \left(\frac{1}{\sqrt{2}} - 1 \right) y = 0$$

$$y = \underline{(\sqrt{2} + 1)z} \quad (ii)$$

$$\Rightarrow p = z \begin{bmatrix} -\sqrt{3} \\ \sqrt{2} + 1 \\ 1 \end{bmatrix}$$

The axis of rotation is $(-\sqrt{3}, \sqrt{2} + 1, 1)$

From lecture slide 5 pg 56

$$\theta = \cos^{-1} \left(\frac{\text{trace}(R) - 1}{2} \right)$$

$$= \cos^{-1} \left(\frac{\frac{1}{2} + \frac{1}{\sqrt{2}} + \frac{1}{2\sqrt{2}} - 1}{2} \right)$$

$$= \cos^{-1} \left(\frac{-\sqrt{2} + 2 + 1}{4\sqrt{2}} \right) = \cos^{-1} \left(\frac{3 - \sqrt{2}}{4\sqrt{2}} \right)$$

$$2) \quad x = K [R | t] X$$

$$[R | t] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \begin{array}{l} \square^{C_1, K_1, n_1} \\ \square^{C_2, K_2, n_2} \end{array} \quad [\text{first frame}]$$

$$\left[\begin{array}{c|c} R & \begin{array}{c} 0 \\ 0 \\ 0 \end{array} \end{array} \right] [\text{second frame}]$$

Since translation is 0 we can drop it from external matrix and Remove the scaling factor so that

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \text{ becomes } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

that is

$$x_i = K_i R X$$

$$x_1 = K_1 I X \quad \text{for } C_1$$

$$x_2 = K_2 R X \quad \text{for } C_2$$

as K is a 3×3 , upper triangular matrix it will be invertible.

$$\text{so } K_1^{-1} x_1 = X \quad H'$$

$$\Rightarrow x_2 = K_2 R K_1^{-1} x_1$$

R is orthogonal 3×3 matrix so it is also invertible

$$\text{we get } x_1 = K_1 R^{-1} K_2^{-1} x_2$$

$$H = K_1 R^{-1} K_2^{-1} \quad \Bigg| \quad H H' = I = H' H$$

as we saw that $H H' = H' H = I$ we can say that H is invertible with H' as its inverse.

3.3 We see that the corners are stretched outwards this is because radial distortion is due to spherical nature of lens which makes it

look like that the edges have been pulled in

3.5 the reprojection error is computed by finding the 3D coordinates of a KeyPoint with the help of the internal and external matrix and after we have computed the 3D KeyPoin

we use that same matrices to compute the point where it would be assuming that there is no distortion

and then taking the gaussian distance over the reprojection and observation gives us reprojection error